

The infinite-sum converse for statistically maximizing sequences fails

Codex agent lane 3 (GPT-5.6)

July 19, 2026

Status and Source Question

Status: counterexample likely valid, pending human review.

Seal–Som–Basu–Dey study farthest-point maximizing sequences and M -compactness in normed spaces [1]. If M is a nonempty bounded subset of a normed space and y is in the ambient space, write

$$\delta(y, M) = \sup_{m \in M} \|y - m\|.$$

A sequence (x_n) in M is *statistically maximizing* when, for some y , the scalar sequence $\|x_n - y\|$ converges statistically to $\delta(y, M)$.

Corollary 3.6 of the source says that statistical maximality in an infinite ℓ_p -sum implies statistical maximality in every coordinate. Remark 3.7 asks whether the converse is true for $1 \leq p < \infty$.

Corollary 3.6. *Suppose $X = \oplus_p X_i$ and $M_i \subseteq X_i$. Let $\{x_n\}_{n \in \mathbb{N}}$ be a statistically maximizing sequence in $\oplus_p M_i$. Then $\{x_n^{(i)}\}_{n \in \mathbb{N}}$ is statistically maximizing in M_i for all $i \in \mathbb{N}$ for $1 \leq p < \infty$.*

Remark 3.7. We do not know whether the converse of Corollary 3.6 is true or not.

The answer is negative for every $1 \leq p < \infty$. The counterexample uses only three-point subsets of two-dimensional Hilbert spaces. In fact, every coordinate sequence below is maximizing in the ordinary sense, while the global sequence is not even statistically maximizing.

Definitions and Construction

For a scalar sequence (r_n) , statistical convergence to r means

$$\lim_{N \rightarrow \infty} \frac{1}{N} |\{n \leq N : |r_n - r| \geq \varepsilon\}| = 0 \quad (\varepsilon > 0).$$

Fix $1 \leq p < \infty$. For $i \in \mathbb{N}$, set

$$\alpha_i = 2^{-i}, \quad X_i = \mathbb{R}^2$$

with the Euclidean norm, and define

$$A_i = \left(-\frac{\alpha_i}{2}, 0\right), \quad B_i = \left(\frac{\alpha_i}{2}, 0\right), \quad C_i = (\alpha_i, \alpha_i^2), \quad M_i = \{A_i, B_i, C_i\}.$$

Let

$$X = \left(\bigoplus_{i=1}^{\infty} X_i \right)_{\ell_p}, \quad M = \left(\bigoplus_{i=1}^{\infty} M_i \right)_{\ell_p} = \{(m_i) \in X : m_i \in M_i \text{ for every } i\}.$$

This is a bounded nonempty subset of X : since $\alpha_i \leq 1/2$,

$$\max_{m \in M_i} \|m\|_2 = \|C_i\|_2 = \alpha_i \sqrt{1 + \alpha_i^2} \leq \frac{\sqrt{5}}{2} \alpha_i,$$

and $(\alpha_i) \in \ell_p$.

Put $A = (A_i)_{i \geq 1}$ and $B = (B_i)_{i \geq 1}$, both elements of M , and define

$$x_n = \begin{cases} A, & n \text{ even,} \\ B, & n \text{ odd.} \end{cases}$$

Proof Intuition

In coordinate i , the two alternating points A_i, B_i are both farthest from the harmless witness $(0, 1)$. Thus each coordinate sequence is as strongly maximizing as possible: its distance to a witness is constantly equal to the farthest distance.

The third point C_i prevents these coordinate witnesses from being replaced by a summable family. If A_i and B_i are both farthest from a center (s_i, t_i) , their equal distances force $s_i = 0$. Comparing either one with C_i then forces

$$t_i \geq \frac{3}{8} + \frac{\alpha_i^2}{2}.$$

Every simultaneous witness therefore has coordinate norm at least $3/8$. Such a family cannot belong to an ℓ_p -sum.

If the alternating global sequence were statistically maximizing, its even and odd distances, each occurring on a set of density $1/2$, would both have to equal the global farthest distance. Coordinatewise additivity of the p th power of the norm would then make every A_i and B_i simultaneously farthest from the corresponding coordinate of one global witness. The lower bound above gives the contradiction.

Formal Counterexample

Lemma 1 (The coordinate obstruction). *For every i , the sequence alternating between A_i and B_i is an ordinary maximizing sequence in M_i . Moreover, if both A_i and B_i are farthest points of M_i from $z_i = (s_i, t_i)$, then*

$$s_i = 0, \quad t_i \geq \frac{3}{8} + \frac{\alpha_i^2}{2}, \quad \|z_i\|_2 \geq \frac{3}{8}.$$

Proof. Take $y_i = (0, 1)$. Direct calculation gives

$$\begin{aligned} \|A_i - y_i\|_2^2 &= \|B_i - y_i\|_2^2 = 1 + \frac{\alpha_i^2}{4}, \\ \|C_i - y_i\|_2^2 &= \alpha_i^2 + (1 - \alpha_i^2)^2 = 1 - \alpha_i^2 + \alpha_i^4. \end{aligned}$$

Their difference is

$$\alpha_i^2 \left(\frac{5}{4} - \alpha_i^2 \right) > 0$$

because $0 < \alpha_i \leq 1/2$. Hence A_i and B_i are both farthest from y_i , and the alternating sequence has constant distance $\delta(y_i, M_i)$.

Now suppose that both A_i and B_i are farthest from $z_i = (s_i, t_i)$. They must be equidistant. Since

$$\|z_i - A_i\|_2^2 - \|z_i - B_i\|_2^2 = 2\alpha_i s_i,$$

we obtain $s_i = 0$. Farthestness relative to C_i gives

$$\begin{aligned} t_i^2 + \frac{\alpha_i^2}{4} &= \|z_i - A_i\|_2^2 \geq \|z_i - C_i\|_2^2 \\ &= \alpha_i^2 + (t_i - \alpha_i^2)^2. \end{aligned}$$

After cancellation and division by $2\alpha_i^2 > 0$, this is exactly

$$t_i \geq \frac{3}{8} + \frac{\alpha_i^2}{2}.$$

The norm bound follows. □

Lemma 2 (Farthest distance in the bounded product). *For every $z = (z_i) \in X$,*

$$\delta(z, M)^p = \sum_{i=1}^{\infty} \delta(z_i, M_i)^p.$$

If $u = (u_i) \in M$ is farthest from z , then each u_i is farthest from z_i in M_i .

Proof. Every M_i is finite, so choose $m_i \in M_i$ attaining $\delta(z_i, M_i)$. The uniform coordinate bound used above implies $(m_i) \in M$. Therefore the coordinatewise upper bound for $\|z - m\|_p^p$ is attained, proving the formula.

If u is globally farthest, then

$$0 = \delta(z, M)^p - \|z - u\|_p^p = \sum_{i=1}^{\infty} (\delta(z_i, M_i)^p - \|z_i - u_i\|_2^p).$$

Every summand is nonnegative, so every summand is zero. □

Theorem 1 (Negative answer to Remark 3.7). *For every $1 \leq p < \infty$, there exist finite bounded sets $M_i \subset \mathbb{R}^2$ and a sequence (x_n) in their bounded ℓ_p -sum such that every coordinate sequence $(x_n^{(i)})$ is maximizing, hence statistically maximizing, in M_i , but (x_n) is not statistically maximizing in the ℓ_p -sum.*

Proof. Use the construction above. Lemma 1 proves that every coordinate sequence is ordinary maximizing.

Suppose, toward a contradiction, that (x_n) is statistically maximizing in M . There is then a $z \in X$ such that

$$\|x_n - z\|_p \xrightarrow{\text{stat}} \delta(z, M).$$

The scalar sequence on the left alternates between the two constants $\|A - z\|_p$ and $\|B - z\|_p$. The even and odd integers both have natural density $1/2$. Statistical convergence is therefore possible only if

$$\|A - z\|_p = \|B - z\|_p = \delta(z, M).$$

Thus both A and B are farthest points of M from z . Lemma 2 implies that, for every i , both A_i and B_i are farthest from z_i in M_i . Lemma 1 now yields

$$\|z_i\|_2 \geq \frac{3}{8} \quad (i \in \mathbb{N}).$$

Consequently $\sum_i \|z_i\|_2^p = \infty$, contradicting $z \in X$. $\square$

Verification Notes

The proof is symbolic and uses no external theorem beyond the definitions in the source. The reusable script `code/check_geometry.py` checks with exact rational arithmetic, for $i = 1, \dots, 40$, both displayed coordinate identities and the equality case at $t_i = 3/8 + \alpha_i^2/2$. It also prints finite-truncation lower bounds on a putative witness norm for $p \in \{1, 1.5, 2, 4, 10\}$. These calculations are a sanity check, not part of the proof.

The most important review points are:

1. the product formula in Lemma 2, including attainment by a coordinatewise selector in the bounded product M ;
2. the density-1/2 argument forcing both alternating global points to be farthest; and
3. the sign and constant in the threshold $t_i \geq 3/8 + \alpha_i^2/2$.

Novelty Audit and Limitations

On July 19, 2026, the local run indexes were searched for arXiv:2008.08491, M -compactness, statistical maximality, and infinite ℓ_p -sums. Web and arXiv searches used the exact phrase from Remark 3.7, the exact source title with citation/year variants, and close variants such as “statistically maximizing” with “direct sum” or “Banach.” They returned the source paper and source-adjacent earlier work on statistical convergence and farthest points, but no later primary source answering this infinite-sum converse. This is a bounded novelty check, not an exhaustive citation audit.

The theorem answers exactly the statistical converse in Remark 3.7 and does so simultaneously for every finite p . It does not concern $p = \infty$, and it makes no assertion about additional structural hypotheses on the sets M_i (for example, convexity) that might restore a converse.

Human-Review Recommendation

Promote as a negative answer to Remark 3.7 if the three review points above check out. The construction is elementary, finite in every coordinate, and the global obstruction is a direct non-summability argument.

References

- [1] S. Seal, S. Som, S. Basu, and L. K. Dey, *Characterization of M -compact sets via statistically convergent sequences*, arXiv:2008.08491v1, 2020.